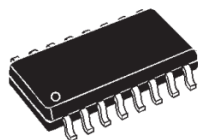


LLC-PFC controller IC for insulated power factor corrector application



SO16N

Features

- Integrated 800 V High-Voltage Startup with high current capability
- Configurable half-bridge controller able to select between half or full bridge operational scheme either with or without topology morphing.
- Adaptive dead-time control
- High accuracy oscillator
- Optimized burst-mode operation at light load with burst mode disable function
- "Safe-start" procedure to prevent hard switching at startup
- Programmable input voltage brown-in/brownout thresholds for single or universal range applications
- Active X-capacitor discharge
- Anti-capacitive mode protection
- 2-level OCP / OVP with autorestart and frequency shift
- IC disable function
- Integrated over temperature protection

Applications

- Insulated PFC pre-regulators IEC61000-3-2 or JEITA-MITI compliant
- SMPS for LED luminaires, LCD TVs, desktop and AIO PCs, servers, Telecom power
- SMPS for USB-PD adapters, single or multiport outputs

Description

The L6790A is an analog IC that can control an insulated PFC converter based on the LLC topology, to comply with the IEC61000-3-2 or JEITA-MITI standards.

L6790A combines resonant topology controller with Power Factor Corrector (PFC) and input current harmonic distortion optimizer (THD). To meet standard requirement and maximize overall performance in all operating conditions, L6790A embeds sensor-less AC input voltage monitor, adaptive dead time and on-the-fly topology morphing control that, when enabled, smoothly and safely swap half bridge mode configuration and full bridge mode configuration according to external operating conditions.

A high efficiency, configurable, burst-mode feature allows improving the efficiency of the converter at light and no load.

A smart High-Voltage Startup Unit provides the required energy to the VCC capacitor during start-up and protection operations.

A full set of protections and their programmability guarantees the correct operation in a broad range of operating conditions.

1 Pin connections and description

Figure 1: L6790A Pinout

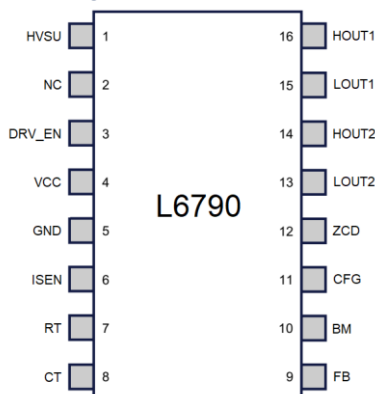


Table 1. Pin description

Name	Pin #	Description
HVSU	1	High voltage start-up generator and input voltage sensing. The pin, able to withstand 800 V, must be connected to the ac side of the input bridge via a pair of high voltage standard rectifier diodes. This pin feeds High voltage startup unit (HVSU), input voltage sensing block and X-cap discharge unit (XCD).
NC	2	Not internally connected. Provision for clearance on the PCB to meet safety requirements.
DRV_EN	3	Driver enable pin. As an output, this pin can be used to enable (active high) the half-bridge drivers and remote circuits to reduce the power consumption during idle mode. As an input the pin can be driven externally to disable the IC (active low).
VCC	4	Supply voltage of the IC. A small bypass capacitor (100 nF typ.) to GND might be useful to get a clean bias voltage for the signal part of the IC.
GND	5	Ground. Current return for the signal part of the IC.
ISEN	6	Negative current sense input. The pin must be connected to sense resistor. It is used for protection purposes (over current protections (OCP1 and OCP2), anti-capacitive-mode (ACM) detection) and adaptive dead time (ADT) control.
RT	7	Minimum frequency programming resistor.
CT	8	Timing capacitor. A capacitor connected from this pin to GND determines the switching frequency of the converter. Its value fixes the maximum switching frequency.
FB	9	Feedback pin. This pin is intended to be directly connected to the collector of the optocoupler from the secondary side to achieve regulation.
BM	10	Burst-mode threshold programming pin. A resistor connected between this pin and GND sets the burst mode threshold. In case the pin is tied to GND at startup or during running, the burst-mode function is disabled. Do not put any capacitor on this pin.
CFG	11	Brownout configuration pin. A resistor connected between this pin and GND sets the brown out level. Do not put any capacitor on this pin.
ZCD	12	Auxiliary voltage sense pin. This pin must be connected to the output of primary side auxiliary winding

		of the transformer by means of a resistor divider to estimate output voltage value.
LOUT2	13	Logic signal for the low-side switch of the 2 nd half-bridge and bridge configuration setup pin.
HOUT2	14	Logic signal for the high-side switch of the 2 nd half-bridge.
LOUT1	15	Logic signal for the low-side switch of the 1 st half-bridge.
HOUT1	16	Logic signal for the high-side switch of the 1 st half-bridge.

2 Block diagram

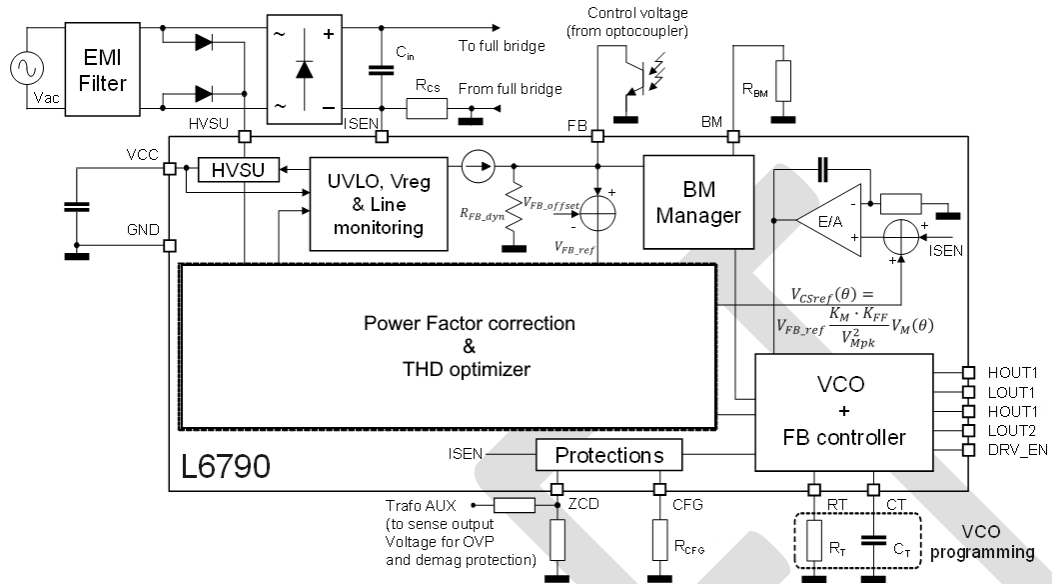


Figure 2: Internal block diagram for L6790A

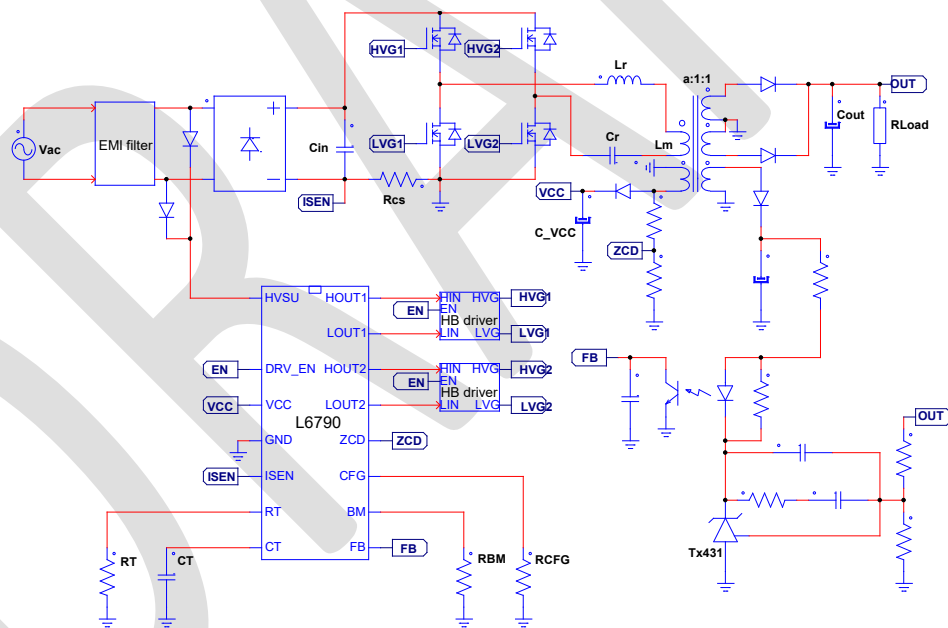


Figure 3: Typical Application schematic L6790A

3 Electrical ratings

3.1 Absolute maximum ratings

Stresses above the absolute maximum ratings listed in Table 1. Absolute maximum ratings may cause permanent damage to the device. These are stress ratings only and functional operation of the device at these conditions is not implied. Exposure to maximum rating conditions for extended periods may affect device reliability.

Table 1: Absolute maximum ratings

Symbol	Pin	Parameter	Value	Unit
V_{HVSU}	HVSU	Voltage range (referred to GND)	-1 to 800	V dc
V_{VCC}	VCC	IC supply voltage	-0.3 to 30	V dc
I_{RT}	RT	Maximum current from RT pin	Self-limited	mA
V_{ISEN}	ISEN	Voltage range	-3 to 3	V dc
$V_{FB}, V_{BM}, V_{CFG}, V_{CT},$	FB, BM, CFG, CT,	Voltage range	-0.3 to 5.3	V dc
$V_{DRV_EN}, V_{LOUT1},$ $V_{LOUT2}, V_{HOUT1}, V_{HOUT2}$	DRV_EN, LOUT1, LOUT2, HOUT1, HOUT2	Voltage range	-0.3 to 5.3	V dc
		$I_{max} < 20mA$ $T_{pulse_max} < 100ns$ not repetitive	6.8	
V_{RT}	RT	Voltage range	-0.3 to 3.3	V dc
V_{ZCD}	ZCD	Voltage range	-TBD to 5	V dc

Where not otherwise indicated the AMRs are intended applied when $VCC > VCC_{PORon}$.

When $VCC < VCC_{PORon}$, the minimum between the indicated value and $VCC+0.3$ V must be considered.

3.2 Recommended operating ranges

When L6790A within recommended operating range, the best performance in terms of control, accuracy and lifetime is obtained.

Table 2: recommended operating range

Symbol	Parameter	Notes	Value			Unit
			Min.	Typ.	Max.	
VCC	Operating voltage	After turn-on	9	-	25	V
$V_{HV,PK_{op}}$	Input voltage peak operative range		105		450	V
$V_{HV_functional}$	Input voltage functional range		0		545	V
F_{line}	Line frequency range		45		66	Hz
R_{CFG}	Configuration programming resistor		15		100	kΩ
R_{RT}	Timing resistor RT operating range		5		30	kΩ
C_{CT}	Timing capacitor CT operating range		270		1000	pF
T_{CT_RT}	CT x RT product operative range		2.5		12	μs
f_{osc}	Oscillator frequency range after T_{OSC_SU}	RT = 10 kΩ, CT = 330 pF	140.8		434.8	kHz
		RT = 20 kΩ, CT = 330 pF	73		387	kHz
		RT = 10 kΩ, CT = 560 pF	85.5		281.3	kHz
		RT = 20 kΩ, CT = 560 pF	43.7		247.7	kHz

3.3 Thermal data

Table 3: Thermal data

Symbol	Parameter	Value	Unit
R_{thj-a}	Maximum junction to ambient thermal resistance	TBD	°C/W
T_J	Junction temperature range	-40 to 150	°C
T_{STG}	Storage temperature range	-55 to 150	°C

3.4 ESD immunity levels

Table 4: ESD immunity levels

Mode	Pin	Reference Specification	Value	Unit
HBM	All except HVSU, OUT, HVG, BOOT	According to JS001	±2	kV
HBM	HVSU, OUT, HVG, BOOT	According to JS001	±1	kV
CDM	All	According to JES002	±500	V

Electrical characteristics

(T_J = -40 °C to 125 °C, 25 °C production tested, V_{CC} = 15 V, C_T = 420 pF, R_T = 23 kΩ, I_{FB} = 150 μA, R_{CFG} = 25 kΩ, R_{BM} = 50 kΩ; DRV_EN, LOUTx and HOUTx not loaded, unless otherwise specified)

Table 5: Electrical Characteristics

Symbol	Parameter	Test condition	Value			Unit
			Min.	Typ.	Max.	
IC supply voltage						
VCCon	Turn-on threshold	Voltage rising ⁽¹⁾		17		V
VCC _{HVSUon}	Charge current turn-on threshold	Voltage falling ⁽¹⁾		12		V
VCCoff	Turn-off threshold	Voltage falling ⁽¹⁾		8		V
VCC _{PORon}	Power-on-reset threshold	Voltage rising ⁽¹⁾		7.5		V
VCC _{PORoff}	Power-on-reset threshold	Voltage falling ⁽¹⁾		7		V
IC supply current						
I _{CC}	Operating supply current			3		mA
I _q	Quiescent current after turn-on	Fault condition		800		μA
I _{BM}	Quiescent current during BM			800		μA
I _{DIS}	Quiescent current in disabled	Pin RT or DRV_EN pulled low		800		μA
High voltage start-up generator and start-up sequence						
V _{HVSU}	Breakdown voltage	I _{HVSU} < 100 μA	800	-	-	V
V _{HVSUstart}	Start voltage (rising)	I _{VCC} < 100 μA; VCC shorted			25	V
V _{VCC_SO}	VCC switchover threshold			1		V
I _{HVSUon}	On-state charge current	Startup V _{VCC} > VCC _{PORon} , V _{HVSU} = 115 Vac		13		mA
		Startup V _{VCC} > VCC _{PORon} , V _{HVSU} = 230 Vac		9		
		Latched condition, restart after timeout or during OTP		4.5		
I _{HVsink}	Off-state current	During brownout and startup phase		100		μA
V _{ZCD_SUend}	IC startup phase end threshold at ZCD pin			1.36		V
T _{HVSU_OUT}	Generator shutdown timeout	After VCC exceeds VCCon		120		ms
I _{HVSUoff}	Off-state leakage current	V _{HV} = 400 V			20	μA
Input voltage sensing						
I _{CFG}	CFG pin current			13.3		μA
		After pinstrap if R _{CFG} < 30 kΩ		26.6		μA
K _{BO/R_{CFG}}	Scaling factor for brownout voltage setting	R _{CFG} from 15 kΩ to 100 kΩ		4		V/kΩ
R _{CFG_MRPH}	R _{CFG} morphing en/dis threshold			47		kΩ
V _{HVPK_BO}	Brownout voltage threshold	R _{CFG} = 25 kΩ		100		V
V _{HVPK_BI}	Brown-in voltage threshold	R _{CFG} = 25 kΩ (in tracking with V _{HVPK_BO})		110		V
V _{HVPK_BOH}	High mains morphing voltage threshold Half Bridge to Full Bridge	R _{CFG} from 15 kΩ to 47 kΩ		235		V

V _{HVPK_BIH}	High mains morphing voltage threshold Full Bridge to Half Bridge	R _{CFG} from 15 kΩ to 47 kΩ (in tracking with V _{HVPK_BOH})		245		V
V _{HVPK_BO_min}	Minimum brownout voltage threshold	R _{CFG} = 15 kΩ		60		V
T _{DB_ACBO}	Brownout debounce time			40		ms
T _{DG_ACBI}	Brown-in deglitch time			500		μs
X-CAP discharge function						
V _{HVSU_min}	Peak residual voltage				45	V
I _{HV_DIS}	Discharge current	V _{HVSU} = 45 V	5			mA
T _{DECT_XCAP}	Detection time			100		ms
Oscillator and adaptive dead time						
V _{CT_ref}	Reference voltage for CT			1.5		V
V _{RT_ref}	Reference voltage for RT			1.5		V
I _{RT_max}	Maximum current from RT pin	During switching	300			μA
I _{EA_max}	Maximum current from E/A			400		μA
I _{EA_min}	Minimum current from E/A	During oscillator startup phase		200		μA
		During normal operation	-5	0	5	μA
I _{OSC_SU}	Oscillator current at startup				412.5	μA
T _{OSC_SU}	Oscillator startup duration			1		ms
n _{OSC_SU}	Oscillator startup steps			16		/
T _{ADT_min}	Minimum dead time	From LOUTx off to HOUTx on		40		ns
T _{ADT_max}	Maximum dead time	From LOUTx off to HOUTx on		420		
T _{pulse_min}	Minimum driving pulse duration before enabling ADT			370		ns
Control logic and outputs						
V _{OL}	LOUTx, HOUTx, DRV_EN "low" voltage				0.4	V
V _{OH}	LOUTx, HOUTx, DRV_EN "high" voltage		4			V
R _{out_DRV_EN}	DRV_EN output resistor			500		Ω
V _{HBFB_th}	LOUT2 Half Bridge – Full Bridge threshold sensing	At startup during safe start		2.5		V
I _{HBFB_th}	LOUT2 current at startup to detect Half Bridge – Full Bridge	At startup during safe start		300		μA
T _{DRV_EN_early}	DRV_EN early time	Before switching turn-on		50		μs
T _{LOUT2_early}	LOUT2 early time	Before turn-on during burst mode		5		μs
Current sense						
V _{CS_pos}	Current sense positive threshold			10		mV
		Maximum threshold during half cycle current sensing		200		mV
V _{CS_neg}	Current sense negative threshold			-50		mV
V _{OCP1}	OCP1 threshold			-550		mV
V _{OCP2}	OCP2 threshold			-750		mV
Feedback and error amplifier						

I _{FB_steady}	FB pin steady state current ⁽²⁾			150		μA
R _{FB_dyn}	FB pin dynamic impedance			35		kΩ
I _{FB_dyn}	FB pin dynamic current	Excluding R _{FB_dyn} consumption	0		367	μA
I _{FB_min}	FB pin minimum current ⁽²⁾			100		μA
I _{FB_dyn_delta}	FB pin dynamic current delta ⁽²⁾			230		μA
V _{FB_gain}	FB gain during run mode			1		/
V _{FB_offset}	Internal offset			0.5		V
V _{FB_range}	FB regulation range amplitude above V _{FB_offset}	During operation	2.7	2.8	2.9	V
V _{FB_open_th}	FB open voltage threshold	Voltage rising (in tracking with V _{FB_offset})		3.5		V
V _{FB_sat_th}	FB saturated threshold	Voltage rising (in tracking with V _{FB_offset})		3.6		V
V _{ISEN_off}	Equivalent voltage offset at ISEN pin	With internal current reference = 0	-2		2	mV
V _{FB_SS}	Internal soft-start duration when no protections are triggered	V _{FB} > V _{FB_open_th}		22.5		ms
		V _{FB} < V _{FB_open_th}		128		ms
Burst mode						
I _{BM}	BM pin current			13.3		μA
		After pinstrap if R _{BM} < 30 kΩ		26.6		μA
V _{BM_eq}	BM threshold on FB voltage	R _{BM} from 15 kΩ to 140 kΩ	0.65		1.9	V
V _{BM_exit_h_eq}	BM exit threshold hysteresis	voltage rising		300		mV
V _{dBM_eq}	Deep BM threshold on FB pin	voltage falling		0.5		V
V _{dBM_h_eq}	Deep BM hysteresis on FB pin			60		mV
T _{BM_res}	Resuming time from BM sleep			60		μs
T _{BM_min}	Minimum switching burst duration	During burst mode		200		
		During VCC recharge		50		μs
V _{BM_gain}	FB gain during BM			1.25		/
V _{BM_dis}	BM disable threshold on BM pin	Voltage falling		0.2		V
Protections						
T _{FB_open}	FB open timeout	FB > V _{FB_open_th} or ZCD > V _{OVP1}		1		s
T _{FB_restart}	Restart time after FB open fault			1		s
T _{UVLO_restart}	Restart time after UVLO fault			2		s
V _{ZCD_pos}	ZCD positive threshold	Voltage rising		0.2		V
V _{ZCD_neg}	ZCD negative threshold	Voltage falling		-0.2		V
T _{ZCD_open}	ZCD open/short fault timeout	ZCD < V _{ZCD}		14		ms
T _{ZCD_restart}	Restart time after ZCD fault	V _{FB} > V _{FB_open_th}		1		s
V _{OVP1}	OVP1 threshold	ZCD pin		2.3		V
V _{OVP2}	OVP2 threshold	ZCD pin		2.5		V
T _{OVP_filt}	OVP1, OVP2 filtering	ZCD pin		500		ns
T _{OVP2_restart}	Restart time after OVP2			100		ms
V _{High_HV}	Input over voltage threshold	HVSU pin		590		V

$T_{\text{fault_restart}}$	Hard ACP/OCP2 fault restart timing after event			50		μs
$V_{\text{IC_dis_DRV}}$	IC disable threshold	DRV_EN pin		1.5		V
$V_{\text{IC_dis_RT}}$	IC disable threshold	RT pin		0.5		V
$V_{\text{IC_dis_h}}$	IC disable hysteresis	DRV_EN and RT pins – Voltage rising		100		mV
$T_{\text{IC_dis_restart}}$	IC disable restart time	DRV_EN and RT pins		2		ms
$T_{\text{IC_dis_samp}}$	IC disable sampling time	DRV_EN and RT pins		15		μs
$I_{\text{RT_open}}$	RT pin open current threshold	RT pin		20		μA
T_{OTP}	HVSU over temperature protection	Temperature rising		165		$^{\circ}\text{C}$
$T_{\text{OTP_h}}$	HVSU over temperature protection hysteresis	Temperature decreasing		50		$^{\circ}\text{C}$

1. Parameters tracking each other

2. Including RFB_dyn consumption

5 Application information

5.1 L6790A typical operation

L6790A is an integrated circuit capable of controlling a single stage resonant converter, providing an isolated voltage directly from the AC mains, from which it draws power with a power factor equal to 1 and input current harmonic distortion close to zero. To do so, the L6790A controller operates an LLC resonant tank shaping its input current accordingly with input voltage.

In its typical configuration it operates over the universal ac mains range re-configuring, on-the-fly, the converter topology from half-bridge to full-bridge and vice-versa (morphing) according to the amplitude input mains voltage.

Nevertheless, the IC can be easily configured to operate without the topology morphing, locking the desired configuration of the bridges (half bridge or full bridge). The typical application is shown in Figure 3.

5.1.1 HVSU pin and power up sequence

The high voltage start up unit, internally connected to HVSU pin is used to charge the VCC capacitor every time the VCC has an insufficient voltage level, in particular at power up and during protection algorithm executions. Figure 4 shows the typical power up sequence.

At power up, during VCC charging, the configuration setup is detected (pinstrap).

At the end of this phase a safe start procedure is initiated to smoothly build up the power transfer to output load. The power up procedure lasts for a maximum period of 120 ms during which L6790A monitors operation parameters to evolve to run mode as soon as control loop is closed. At the end of this timer, the L6790A enables the run mode operation regardless the control loop stability is reached.

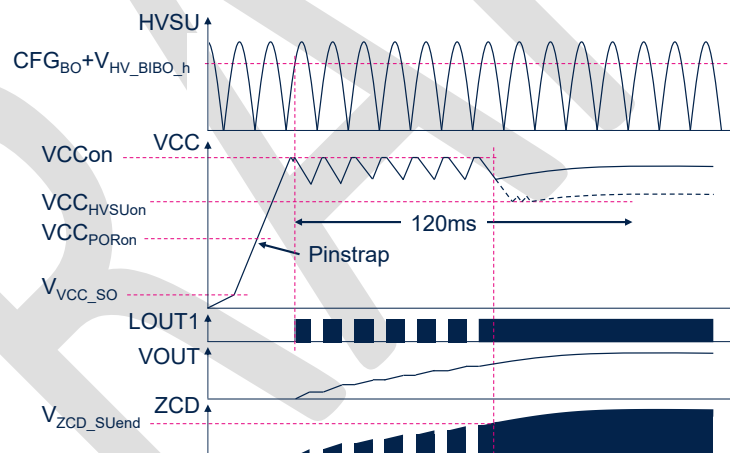


Figure 4: power-up sequence

5.1.2 Run mode

When the startup phase ends, the L6790A enters the run mode. In this phase the FB pin voltage, driven by the output of the error amplifier isolated by an optocoupler, sets the amount of transferred power.

The oscillator, configured by RT and CT pair, generates the resulting operating frequency having a distribution controlled by power factor corrector block.

The switching frequency is managed by PFC and THD optimizer block that generates driving pattern for the switch outputs considering the configuration (half bridge / full bridge) set by programmed configuration and input voltage.

5.1.3 Topology morphing

To cover the universal mains range, the L6790A can toggle (morph) the bridges topology on-the-fly, to permit a proper resonant tank operation over wide range of the input voltage. When the peak mains voltage is below the VHVPK_BOH threshold, the L6790A silently, with no perceivable interruptions,

moves from half-bridge to full-bridge configuration and vice versa.

Precise and noise-free peak detector block ensures a stable transition as well as a correct configuration during power up phase.

5.1.4 Adaptive dead time feature

The selection of the proper dead-time represents a key element to maximizing the efficiency of any resonant converter topology.

L6790A embeds an adaptive dead time algorithm able to optimize the operating dead time of the running half bridges.

5.1.5 Burst mode

The burst mode feature allows to reduce the total power consumption, increasing the conversion efficiency.

When the power level set by the FB pin is below the burst mode (BM) entering threshold set by RBM, the L6790A stops the switching activity, reducing power consumption, in correspondence of the mains zero crossing: this operating mode is maintained for at least 8 half mains cycles.

The continuous run mode is activated if FB voltage stays for at least 8 consecutive half mains cycle above the burst mode entering threshold.

The run mode activation is forced if the upward increase of FB is larger than $V_{BM_exit_h_eq}$ (large load variations).

The driver signal pattern at burst mode exit is executed according to the switch configuration: this approach ensures a safe bootstrap charging of the floating driver of each half bridge.

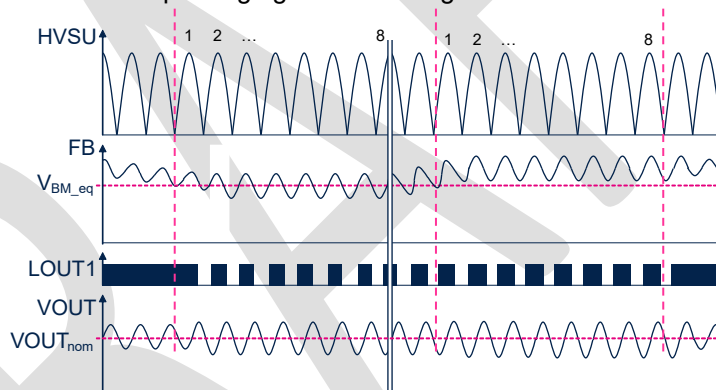


Figure 5: Burst mode management with L6790A

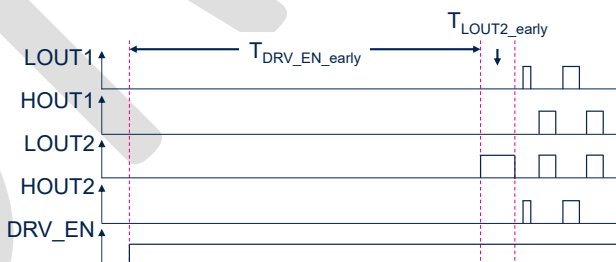


Figure 6: Burst mode restart in Full Bridge configuration

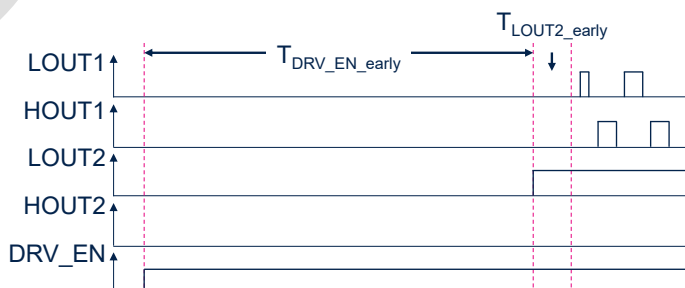


Figure 7: Burst mode restart in Half Bridge configuration

A second, lower, FB threshold activates, when crossed downwards, the deep burst mode (dBm): at this occurrence L6790A stops switching immediately and restores the burst mode operation as soon as VdBM_eq threshold is crossed upwards.

The burst mode functionality can be disabled if the BM pin voltage is put below the VBM_dis threshold: in this case the IC will never enter burst mode, while deep burst mode is always active and run mode is immediately restored at the exit of deep burst mode.

5.1.6 Disable mode

The IC can be remotely disabled pulling either the DRV_EN and pins the RT to GND.

When disabled, the L6790A stops switching and minimizes the VCC consumption. The high-voltage startup is turned on at reduced current capability to keep the IC supplied.

The DRV_EN and the RT pins state is checked every 2 ms: if the disabling condition is removed, L6790A restarts switching with safe start.

In case the brownout fault is triggered during disable, the periodical disable pin check is interrupted until the IC detects the ac brown-in condition.

To avoid false triggering of the disable mode, avoid putting external capacitor on the DRV_EN and RT pins.

5.1.7 X-Cap Discharge

L6790A embeds X-Cap discharge feature. It discharges the EMI filter capacitors connected between the phase and the neutral wire (X type caps) to a safe voltage level when the mains voltage is removed be compliant with the safety regulations. HVSU unit is activated to realize this function while smart filtering is provided to avoid false triggering.

5.1.8 L6790A supply pin

The VCC pin is used to supply the IC. High voltage start-up unit supplies the VCC at power up and during inactive modes, while the auxiliary winding voltage can be used to supply the VCC pin during normal operation.

When additional voltage regulators are used to limit the VCC voltage within operating conditions, a bypass diode shall be put between the regulator output and the VCC pin to decouple HVSU charging current.

5.2 Protections

5.2.1 UVLO

L6790A embeds an UVLO protection that stops the switching activity in case of insufficient VCC voltage. The switching activity restarts with safe start procedure after a minimum inactive period of 2 seconds and during which HVSU maintains VCC capacitor charged.

5.2.2 Brownout fault and brown-in threshold

The brownout fault stops the switching activity in case the instantaneous input voltage is below the brownout threshold for TDB_ACBO (40 ms typ.).

As soon as the input voltage is above the brown-in threshold, the HVSU recharges the VCC capacitor and the switching activity starts with safe start procedure.

5.2.3 Demagnetization sensing

The demagnetization protection is designed to avoid hard switching during the first pulses.

Every time the L6790A starts switching, the pattern generator can increase the duration of turn-on pulse or dead-time according to the voltage and current of the resonant tank coming from the ZCD and ISEN pins (see Figure 3).

Protection is realized also taking into account the running configuration of the topology (half bridge or full bridge).

This protection is automatically put on hold as soon as the adaptive dead-time is enabled and ready to be invoked again as soon as a new restart procedure is realized.

5.2.4 Internal soft-start

The L6790A provides a maximum power control feature using an internal structure that realizes two main features: a soft-start procedure that progressively increase the maximum power delivery at start up and a power limitation when some protections are triggered; in particular, when ACP soft, OCP1 or OVP1 the structure limits the output power restarting the soft start slope.

5.2.5 Anti-capacitive protections

The anti-capacitive protections prevent the resonant tank from switching into capacitive region, resulting in hard switching of the two half bridges.

There are two levels of protection: the ACP-soft, that increases the switching frequency limiting the output power, and the ACP-hard, that stops the switching activity for 50 μ s before new restart at the maximum switching frequency.

Both ACPs are internally connected to the ISEN pin (see Figure 3)..

5.2.6 Over current protections

L6790A embeds two over current protections to limit the current on the resonant tank, sensing the voltage on the ISEN pin (see Figure 3).

The 1st level OCP (also known as OCP1), when triggered, the L6790A rapidly increases the switching frequency charging the internal CSS capacitor.

The 2nd level OCP (also known as OCP2), when triggered, the L6790A stops immediately the switching activity for 50 μ s and performs new restart attempt at the startup frequency.

Both overcurrent protections are blanked in correspondence of the gate turn-on.

5.2.7 Output over voltage protections

L6790A embeds two levels of output over voltage protection on ZCD pin (see Figure 3).

The 1st level OVP (also known as OVP1), when triggered (if the ZCD pin goes above $VOVP1 = 2.3$ V typ.), continues the switching activity with reduced power throughput, recharging the internal CSS capacitor.

The 2nd level OVP (also known as OVP2), when triggered ($VOVP2 = 2.5$ V typ.), stops the switching activity for 100 ms (i.e. $TOVP2_restart$) before restarting with the safe start.

5.2.8 ZCD disconnection

The L6790A includes a ZCD disconnection protection to avoid operation if nor switching activity or output voltage are sensed. If no switching activity is detected on ZCD pin while the output voltage is regulated, the L6790A stops any activity in a latched mode.

5.2.9 ISEN disconnection

The L6790A includes ISEN shorting protection to avoid operation if nor switching activity or resonant current are sensed for 7 consecutive switching cycles. If no signal is detected on ISEN and ZCD pins, the L6790A stops any activity in a latched mode.

5.2.10 FB disconnection and overload protection

The L6790A includes FB disconnection protection that is triggered in case the FB pin voltage stays above the $V_{FB_open_th}$ threshold for T_{FB_open} (1 s typ.). At this occurrence the system is stopped for $T_{FB_restart}$ (1 s typ.) before attempting new restart procedure.

5.2.11 Input over voltage protection

In case the input voltage is above V_{High_HV} threshold, as in case of surges or lightening, to protect the power switches, the L6790A will stop switching for 50 μ s, then it restarts switching at maximum switching frequency, until the condition disappears.

5.2.12 FB anti-saturation circuit

The IC integrates a patented feature that increases the speed of the feedback response when a large signal transition saturates the standard dynamic range of the feedback itself.

To maximize the linear range of the anti-saturation circuit, it is important to select properly the biasing resistor in series with the optocoupler:

$$R_{B_max} = \frac{V_{supply} - (V_{O_min} + V_F)}{\frac{V_F}{R_p} + \frac{I_{FB_steady}}{CTR_{@I_{FB_steady}}}};$$

$$R_{B_min} = \frac{V_{supply} - (V_{O_min} + V_F)}{\frac{V_F}{R_p} + \frac{I_{FB_max}}{CTR_{@I_{FB_max}}}}$$

where:

- V_{supply} is the supply voltage of the feedback circuit;
- V_{O_min} is the minimum output voltage of the error amplifier (for shunt regulator it is usually the regulation voltage);
- V_F is the forward drop of the photodiode;
- R_p is the resistor put in parallel to the photodiode to supply a minimum current to the error amplifier when open (for shunt regulator);
- I_{FB} and CTR are the primary side feedback current (i.e. the phototransistor current) and respectively current transfer ratio of the optocoupler at steady state current (150 µA typ.) and at the maximum feedback current (300 µA min).

Please note that the CTR given into the datasheet is based on the photodiode current, on which it strongly depends.

5.2.13 HVSU over temperature protection

When an excessive temperature is detected on high voltage startup unit, the charging current capability is reduced to let the L6790A to cool down. The full current capability is restored when the junction temperature of HVSU returns below safe levels.

5.3

Designing with L6790A

5.3.1

L6790A Configuration setup

The resistor connected between CFG pin of L6790A and GND, together with the presence of a strong pull down on LOUT1 set the operating scheme of power switches:

- Morphing (automatic swap between full and half bridge operating scheme according to input voltage)
- Half bridge (in this configuration, split resonant capacitor must be used)
- Full bridge

Contemporarily, the threshold values for Brown-out threshold (BIBO) can be adjusted.

$$V_{BO} = \min(60 V, R_{CFG} \cdot K_{BO/R_{CFG}}) = \min(60 V, R_{CFG} \cdot 4 \frac{V}{k\Omega})$$

No external capacitor must be placed on the CFG pin while RCFG over operating range will latch the IC.

Table 6 summarizes the different configuration schemes and setting values.

Table 6: L6790A configuration with RCFG and LOUT2

Configuration	RCFG	LOUT2	Input peak voltage	Operating mode
Morphing Fixed BIBO thresholds	15 kΩ	OPEN	Vin < 60 V	Stop
			70 V < Vin < 235 V	Full Bridge
			Vin > 245 V	Half Bridge
Morphing Adjusted BIBO thresholds	15 kΩ ... 47 kΩ		Vin < CFG _{BO}	Stop
			CFG _{BO} + 10 V < Vin < 235 V	Full Bridge
			Vin > 245 V	Half Bridge
Fixed full bridge Adjusted BIBO thresholds	47 kΩ .. 100 kΩ		Vin < CFG _{BO}	Stop
			Vin > CFG _{BO} + 10 V	Full Bridge
Fixed half bridge (Split Cr) Fixed BIBO thresholds	15 kΩ	GND	Vin < 60 V	Stop
			Vin > 70 V	Half Bridge
Fixed half bridge (Split Cr) Adjusted BIBO thresholds	15 kΩ .. 100 kΩ		Vin < CFG _{BO}	Stop
			Vin > CFG _{BO} + 10 V	Half Bridge

5.3.2

Switching frequency and oscillator components selection

The value of switching frequency in different operating phases is defined by the activity of the oscillator that is programmed by the pair RT and CT. A simplified diagram of this block is illustrated in picture below, where CT charging currents are indicated.

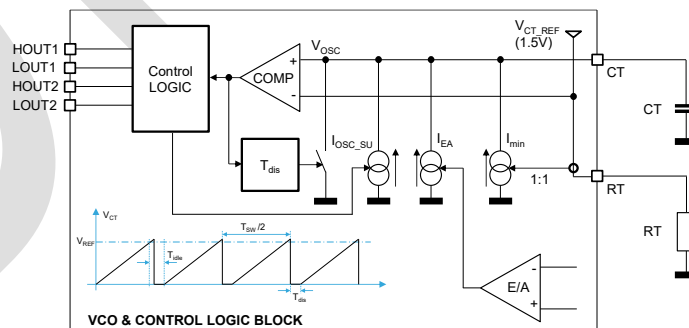


Figure 8: Oscillator structure

The maximum value of CT is defined considering the desired range of operating frequencies using the following formula:

$$CT_{max} = 133.34 \frac{\mu A}{V} \cdot \frac{(700ns \cdot f_{sw_max} - 1) \cdot (700ns \cdot f_{sw_min} - 1)}{f_{sw_max} - f_{sw_min}}$$

Once desired CT is obtained, the minimum RT value to obtain minimum switching frequency can be found using following formula:

$$RT_{min} = \frac{1}{CT} \cdot \left(\frac{1}{2 \cdot f_{sw_min}} - 700ns \right)$$

With the selected couple of CT and RT, the minimum, the maximum and the startup frequencies can be calculated as:

$$f_{sw_min} = \frac{1}{2 \cdot (CT \cdot RT + 700ns)}$$

$$f_{sw_max} = \frac{1}{2 \cdot \left(\frac{CT}{266.67 \frac{\mu A}{V} + \frac{1}{RT}} + 700ns \right)}$$

$$f_{sw_SU} = \frac{1}{2 \cdot \left(\frac{CT}{541.34 \frac{\mu A}{V} + \frac{1}{RT}} + 700ns \right)}$$

In case the startup frequency is too low, is it possible to increase it by reducing the CT capacitor and re-evaluating the RT resistor, though considering that the minimum capacitor must be higher than:

$$CT_{min} = \frac{1}{30k\Omega} \cdot \left(\frac{1}{2 \cdot f_{sw_min}} - 700ns \right)$$

The maximum operating frequency is upper limited to about 675 kHz.

5.3.3 Maximum power control and current sense selection

The IC embeds a maximum power control feature that limits the maximum deliverable output power according to the value of current sense resistor (RCS).

$$R_{CS_max} = \frac{16.8 \Omega \cdot W}{Pin_max}$$

RCS value sets the OCP1 protection trigger point: the lower between the two values should be used for actual RCS value. Do not put any filter or resistor from the current sense resistor to the ISEN pin to alter either maximum power control or overcurrent protection setpoint.

5.3.4 ZCD pin connection

The ZCD pin is used to monitor the output voltage and to avoid hard switching during the first switching pulses.

For correct operation, the auxiliary winding of the transformer must be connected in order to obtain positive voltage on ZCD pin when the LOUT1 (and eventually HOUT2) is on.

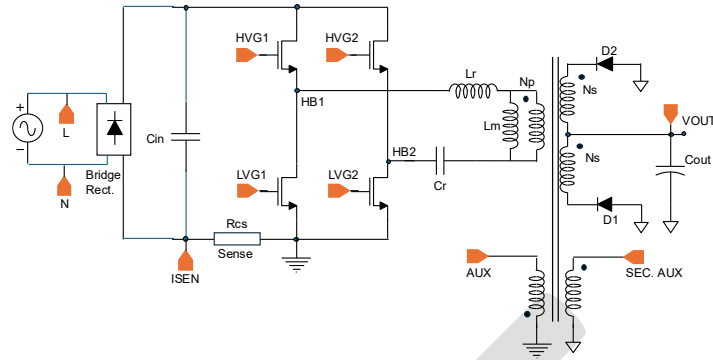


Figure 9: Resonant transformer connection

The selection of the ZCD resistors must fulfill the following equations:

$$R_{ZCD_L} = \frac{V_{out_OVP1}}{V_{OVP1}} \cdot \frac{n_{aux_pri}}{n_{sec}} \cdot \frac{1}{I_{bias}}$$

$$R_{ZCD_H} = R_{ZCD_L} \cdot \frac{V_{out_OVP1}}{V_{OVP1}} \cdot \frac{n_{aux_pri}}{n_{sec}} - R_{ZCD_L}$$

where I_{bias} is the biasing current of the resistor divider at the OVP1 threshold (100 μA is suggested) and V_{out_OVP1} is the maximum output voltage where the OVP1 protection takes control. The relation between output voltage and ZCD voltage is reported in the following relation.

$$V_{out} = V_{ZCD} \cdot \left(\frac{R_{ZCD_H}}{R_{ZCD_L}} + 1 \right) \cdot \frac{n_{sec}}{n_{aux_pri}}$$

5.3.5 BM pin configuration and setting

As described in 5.1.5 BM pin is used to either adjust burst mode entry point or to disable the burst mode functionality. In first case, a resistor placed between BM pin and GND, is selected accordingly the following formula:

$$R_{BM} = 16.7 \frac{k\Omega}{V^2} (R_{CS} \cdot P_{in_BM})$$

The selected RBM is read during power up and fixes the equivalent burst mode threshold on FB voltage. This threshold can be calculated as follows and is summarized in the graph below:

$$V_{BM_eq} = \frac{R_{BM}}{100k} + 0.5V$$

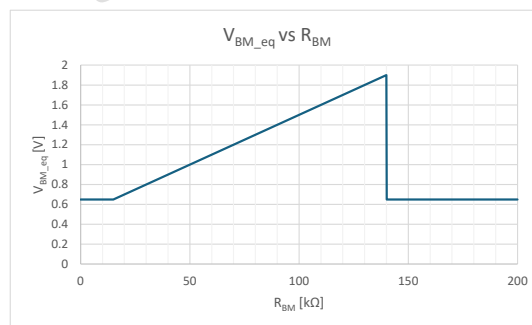


Figure 10: Burst mode threshold vs RBM

If the BM pin voltage is put below the V_{BM_dis} threshold the burst mode functionality is disabled. To avoid unwanted behavior, do not put any external capacitor on the pin.

6 Package information

In order to meet environmental requirements, ST offers these devices in different grades of ECOPACK packages, depending on their level of environmental compliance. ECOPACK specifications, grade definitions and product status are available at: www.st.com. ECOPACK is an ST trademark.

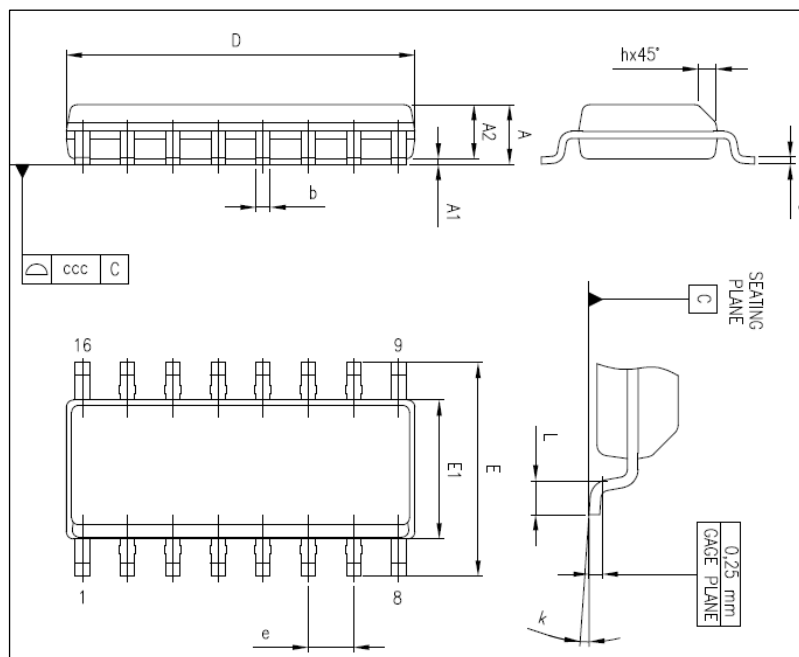


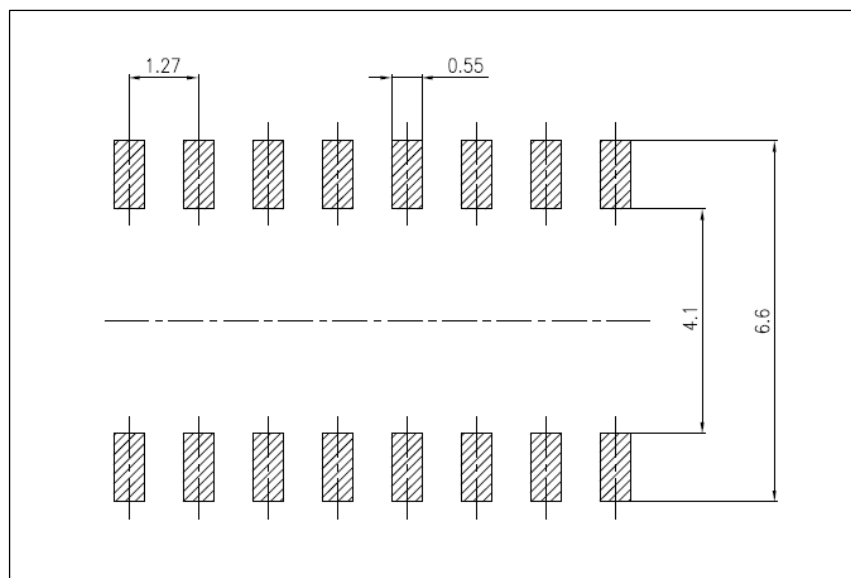
Figure 11. SO16N package outline

Table 7. Mechanical data

Symbol	Dimensions [mm]			Note
	Min.	Typ.	Max.	
A	1.35	-	1.75	
A1	0.10	-	0.25	(2)
b	0.35	-	0.49	
c	0.19	-	0.25	
D	9.80		10.00	(1)
E	5.80		6.20	
E1	3.80		4.00	(1)
e		1.27		
L	0.40		1.25	
h	0.25	-	0.50	
k	0		7	(3)
ccc		0.10		

- (1) Dimensions D and E1 do not include mold protrusion. Max mold protrusion 0.15mm per side.
 (2) (2) Dimension b does not include DAM BAR protrusion. Allowable DAM BAR protrusion shall be 0.127mm total in excess of the b dimension at maximum material condition.
 (3) Degrees.

Figure 12. suggested footprint



9 Ordering information

DRAFT

Revision history

Table x. Document revision history

Date	Revision	Changes
	1	First release



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